

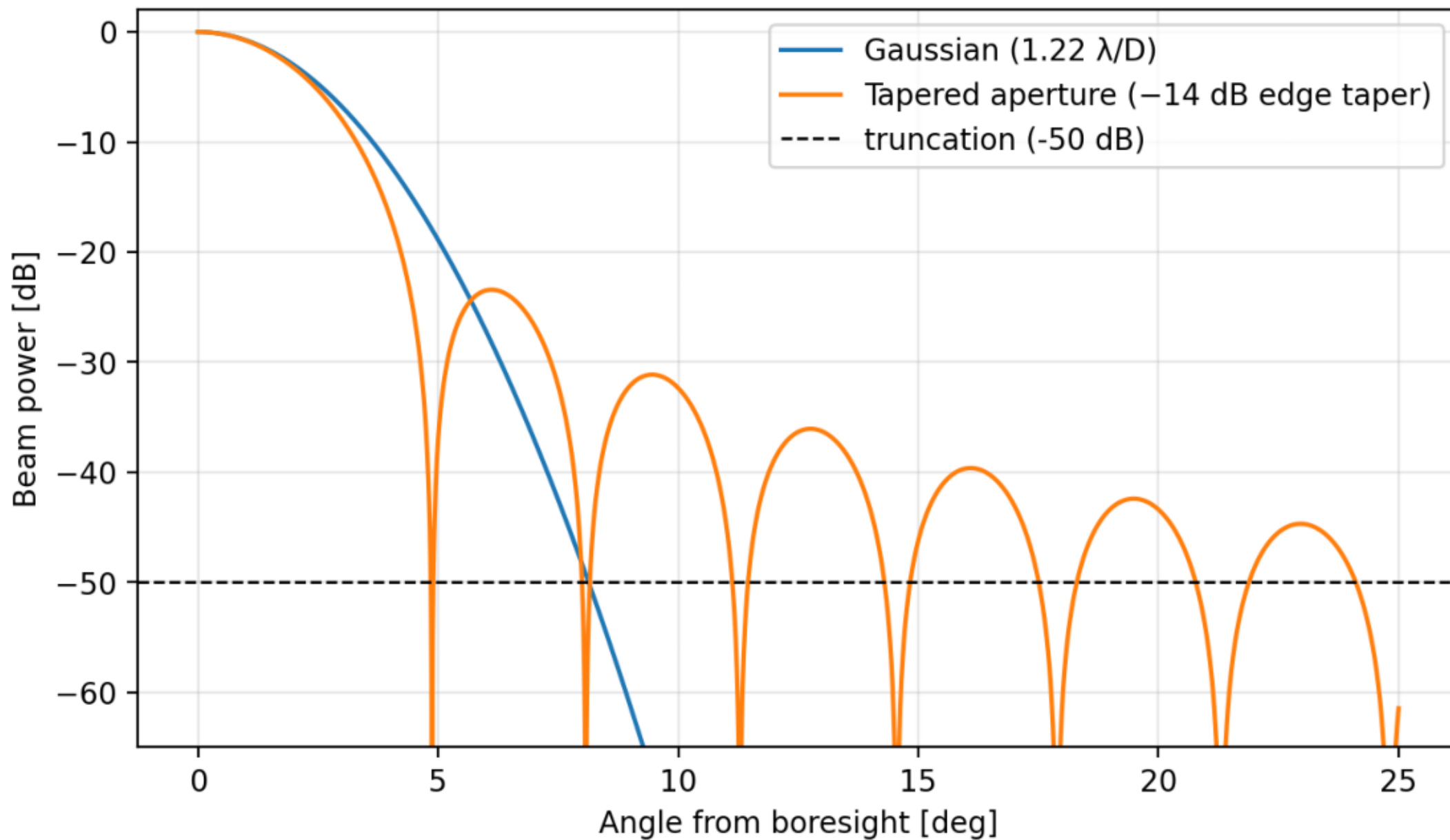
Experiment 001 — short-drift-matched-beams

2026-07-24

Matched forward/inverse beams: Gaussian (FWHM 4.0°) vs tapered aperture (FWHM 3.8° , first sidelobe -23 dB). Short drift scan at 350 MHz: SKA-Mid-like 15 m dish at the Karoo site, parked at $az = 0^\circ$, elevations [52.0, 50.0, 48.0] on 3 consecutive sidereal nights (1 h/night, $dt = 2$ s, 5400 samples), GDSM sky at $n_{side} = 64$, default $1/f$ gain noise + fractional white noise $2.5e-06$, per-night seeds 0+night shared across beams. Wiener map-maker with beam-smoothed-truth prior; solved without HP filter and with a 2 MHz cutoff.

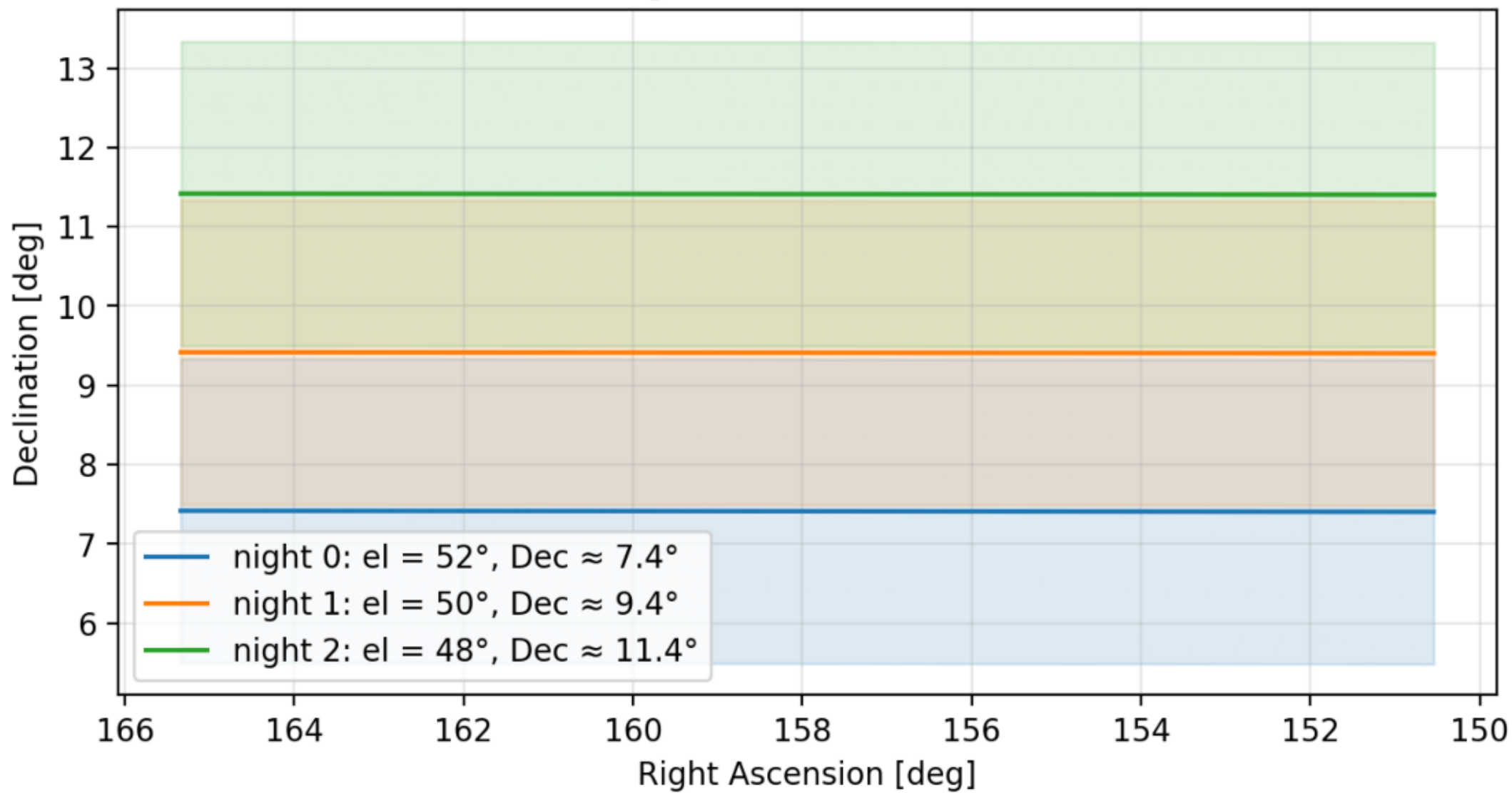
gauss / no HP	1299.9 mK
gauss / HP 2 MHz	1173.6 mK
airy / no HP	1227.3 mK
airy / HP 2 MHz	1090.8 mK

SKA-Mid-like 15 m dish beam models at 350 MHz

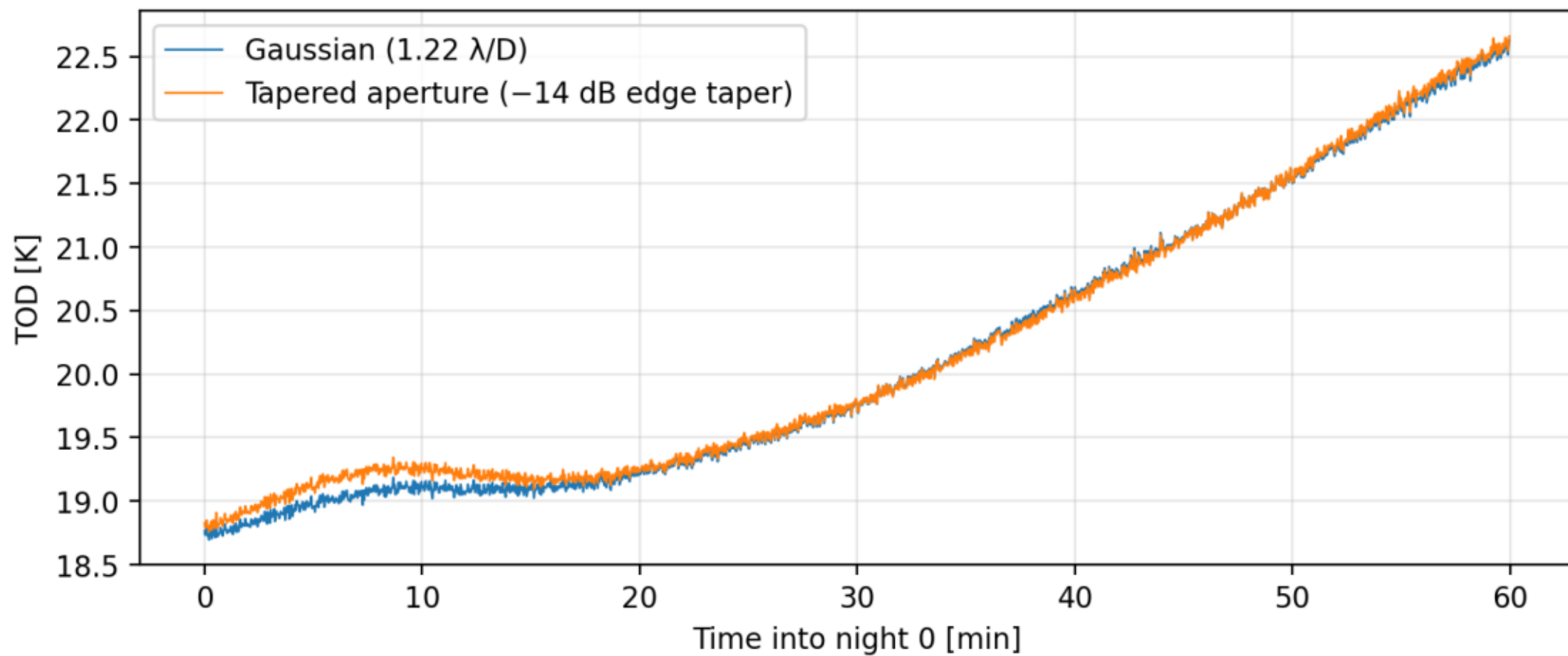


drift_tracks.png

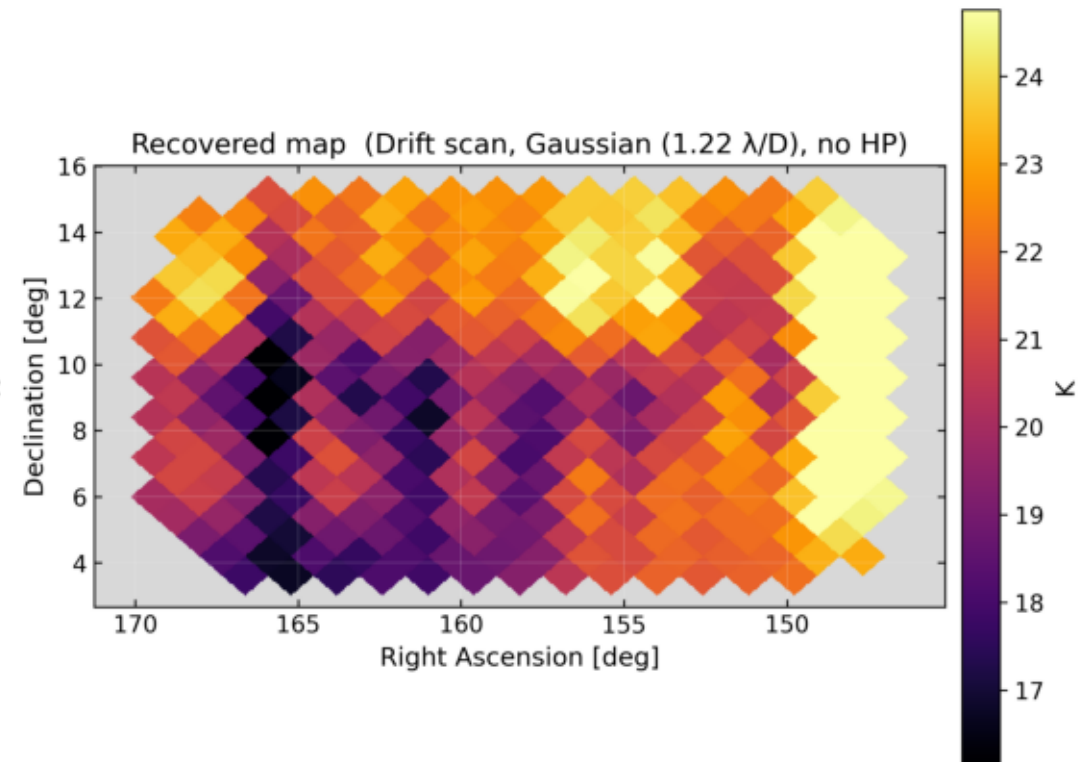
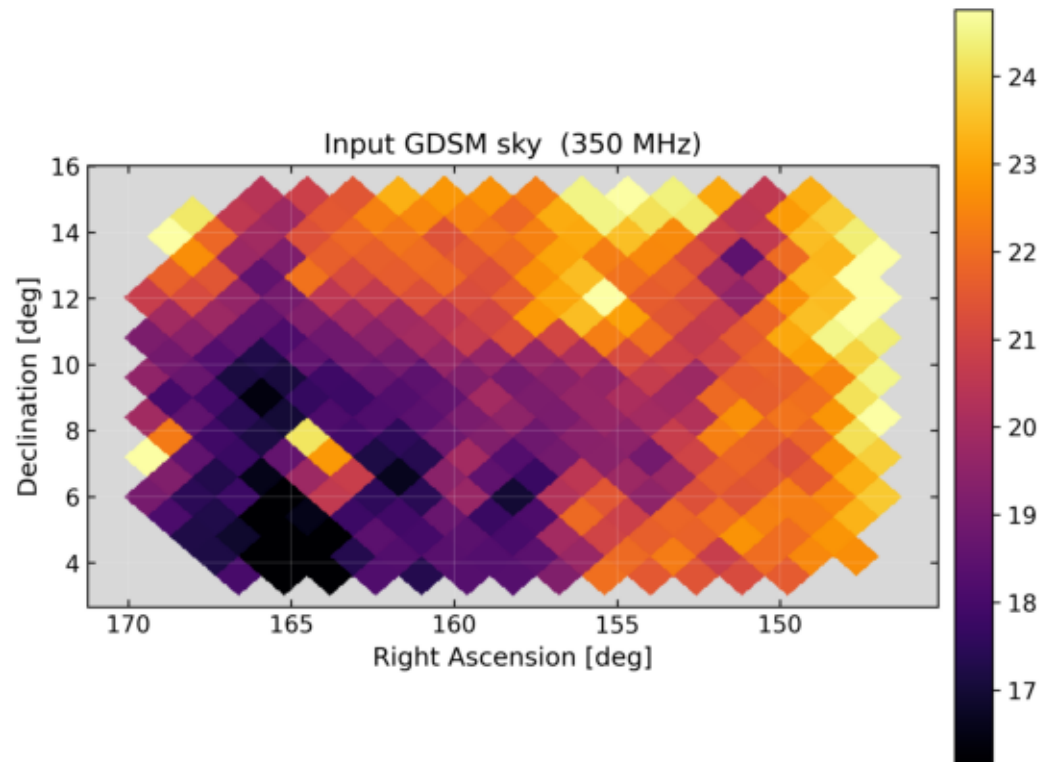
Drift-scan boresight tracks (bands = beam FWHM)



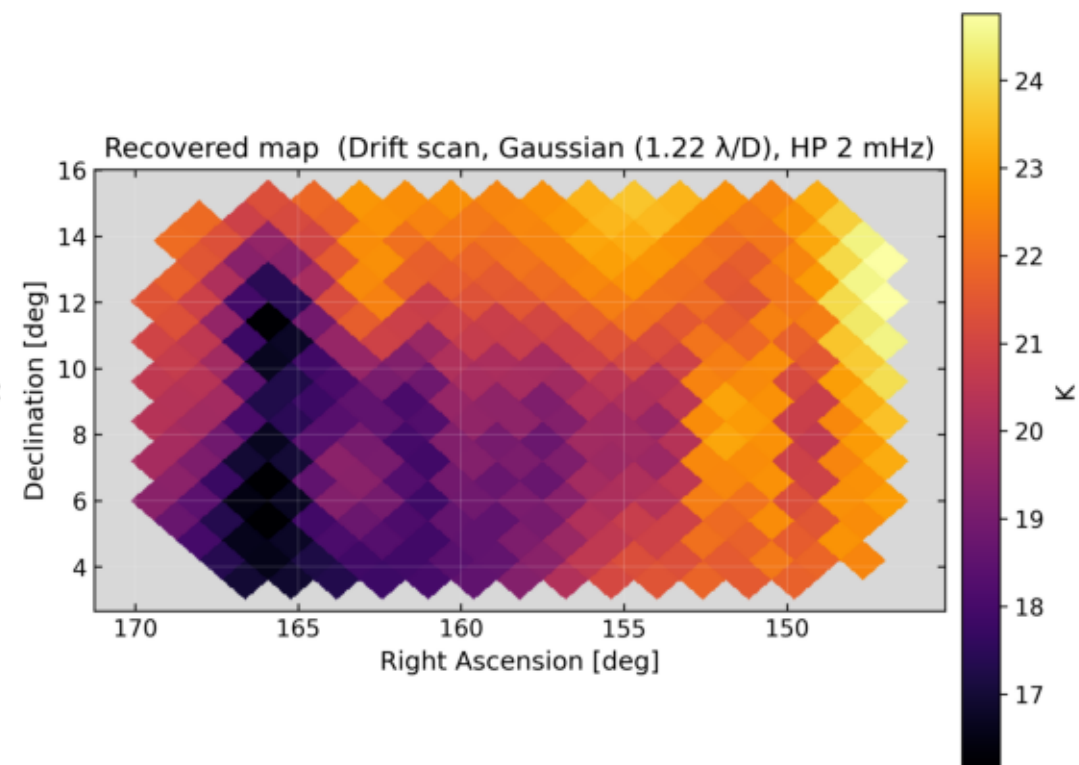
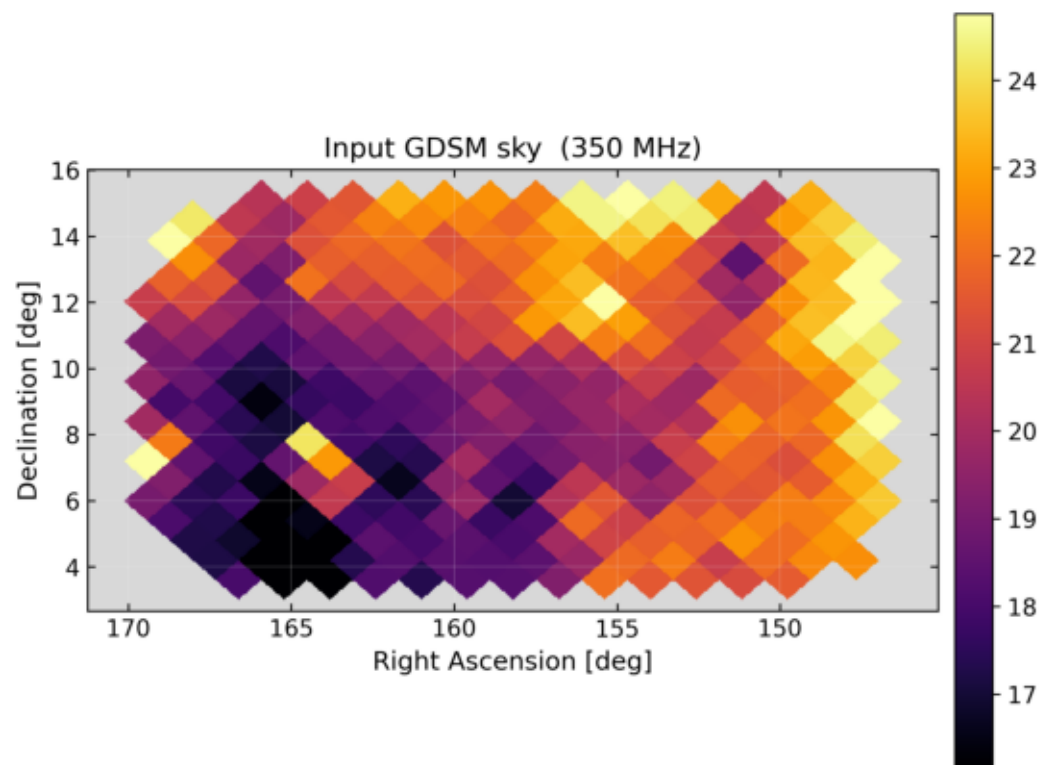
tod_night0.png



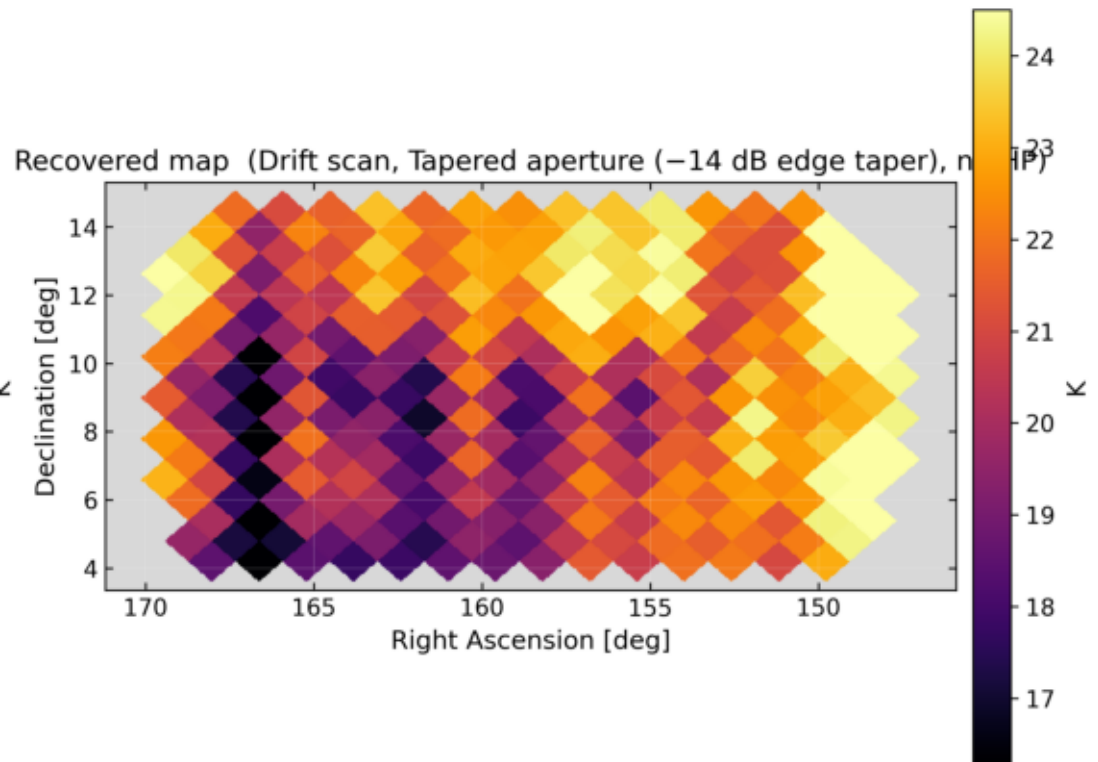
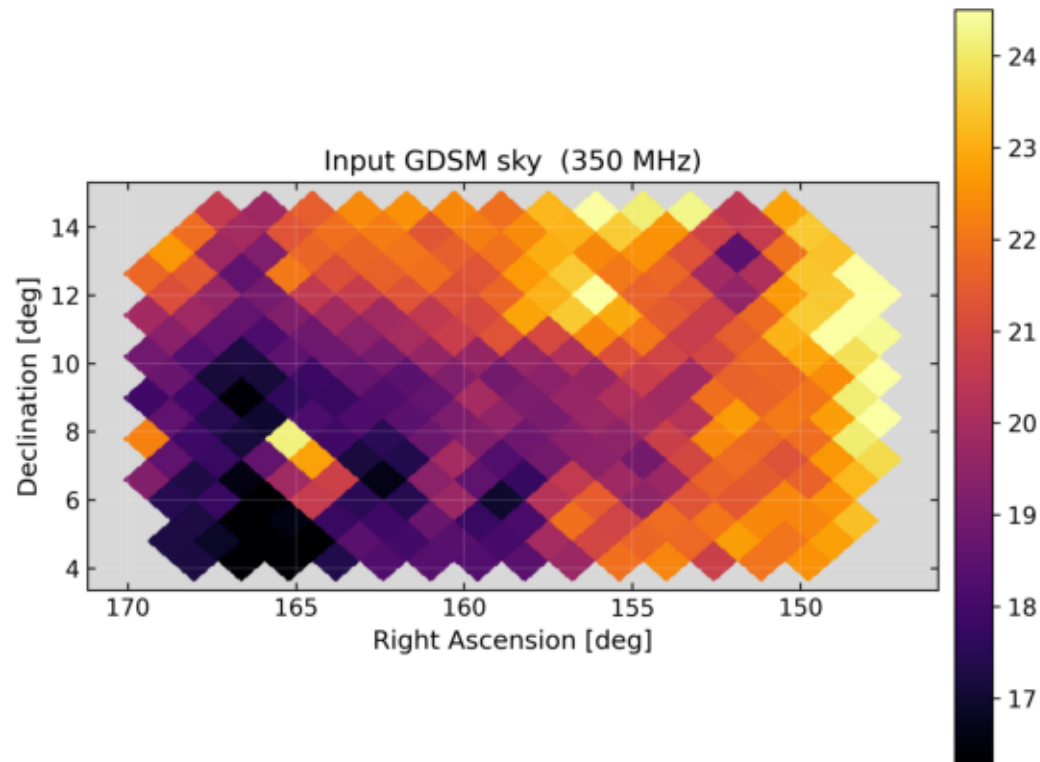
map_ska_drift_gauss_noHP.png



map_ska_drift_gauss_HP.png



map_ska_drift_airy_noHP.png



map_ska_drift_airy_HP.png

